Separable Equations

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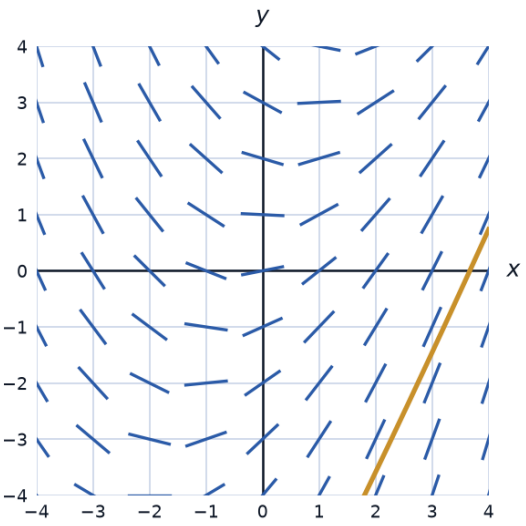
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Order and verifying solutions

State the order of the differential equation $y'' + 3y' - 2y = \sin x$.

2 [Foundational] • Target: Order and verifying solutions

State the order of $\left(\frac{dy}{dx}\right)^3 + xy = 0$, and note that order depends only on the highest derivative, not on any power of it.



3 [Foundational] • Target: Order and verifying solutions

Verify that $y = e^{2x}$ is a solution of $y' - 2y = 0$.

4 [Foundational] • Target: Equilibrium solutions and qualitative behavior

Find the equilibrium (constant) solutions of $\frac{dy}{dx} = y(3 - y)$.

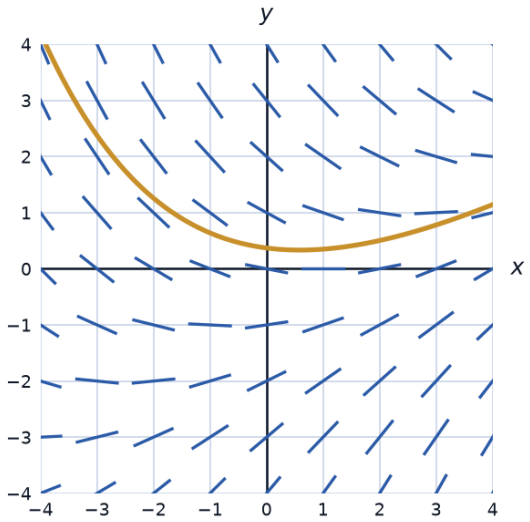
5 [Foundational] • Target: Equilibrium solutions and qualitative behavior

For $\frac{dy}{dx} = y(3 - y)$, determine whether $y = 0$ and $y = 3$ are stable or unstable by checking the sign of $\frac{dy}{dx}$ just above and below each.

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6 [Foundational] • Target: Equilibrium solutions and qualitative behavior

Find the equilibrium solutions of $\frac{dy}{dx} = (y-2)(y+1)$ and classify each as stable or unstable.



7 [Foundational] • Target: General versus particular solutions

Find the general solution of $\frac{dy}{dx} = 2x$, then find the particular solution satisfying $y(1) = 3$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: General versus particular solutions

Explain, in terms of the arbitrary constant C, why a differential equation typically has infinitely many solutions but an initial value problem typically has exactly one.

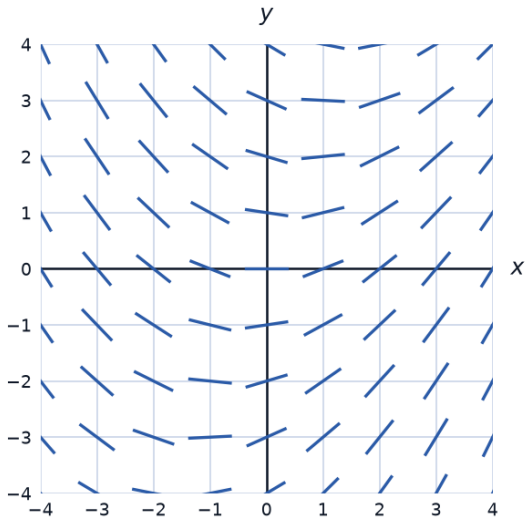
9 [Intermediate] • Target: Application: radioactive decay and population growth

A radioactive substance decays according to $\frac{dA}{dt} = -kA$. If $A(0) = 100$ g and $A(5) = 50$ g, find k.

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10 **[Intermediate]** • **Target: Application: radioactive decay and population growth**

A population grows according to $\frac{dP}{dt} = 0.03P$, with $P(0) = 200$. Find $P(10)$.



11 **[Intermediate]** • **Target: Application: radioactive decay and population growth**

A bank account earns continuously-compounded interest, $\frac{dA}{dt} = 0.05A$, with $A(0) = 1000$. Find $A(20)$ and the doubling time.

12 **[Intermediate]** • **Target: Application: Newton's Law of Cooling**

Newton's Law of Cooling gives $\frac{dT}{dt} = -k(T - T_s)$, where T_s is the surroundings' temperature. Separate variables and integrate to find the general solution.

13 **[Intermediate]** • **Target: Application: Newton's Law of Cooling**

A cup of coffee at $90\,^\circ$C is left in a $20\,^\circ$C room. If $T(10) = 60\,^\circ$C, find k, then find $T(20)$.

14 **[Intermediate]** • **Target: Mixed review**

Determine whether $\frac{dy}{dx} = x + y$ is separable. Justify your answer.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

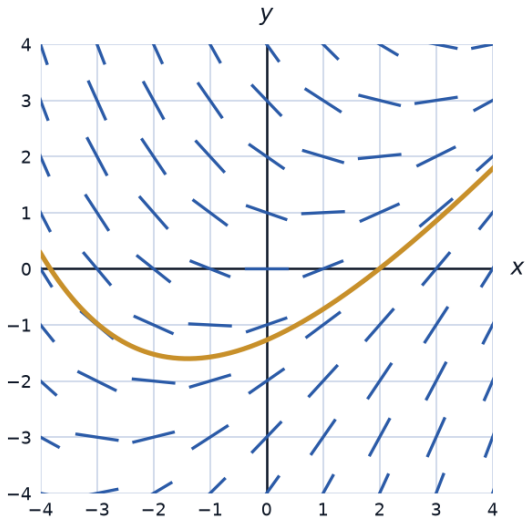
15 **[Challenging]** • **Target: Mixed review** • **Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients**

A tank of water drains according to Torricelli's Law, $\frac{dh}{dt} = -kh^{1/2}$, where h is the water height. Separate variables and integrate.

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16 [Challenging] • Target: Mixed review • Cross-Lesson: 5.01 Variation of Parameters

For the model $\frac{dy}{dx} = \frac{2x}{y}$ with $y(0) = 3$, find the particular solution and state its domain.



17 [Challenging] • Target: Further practice: separable equations • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Find all equilibrium solutions to $\frac{dy}{dx} = y(4 - y) - 3$, classify their stability, and find the particular solution passing through $y(0) = 2$.

18 [Challenging] • Target: Mixed review • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

Compute the first three Picard successive approximations $y_1(x), y_2(x), y_3(x)$ for the initial value problem $\frac{dy}{dx} = 2x(1 + y)$ with $y(0) = 0$, and identify the Taylor series of the exact solution.

19 [Challenging] • Target: Order and verifying solutions • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

Solve the initial value problem $\frac{dy}{dx} = 2x(y^2 + 1)$ with $y(0) = 1$, and determine its maximal interval of existence.

20 [Challenging] • Target: Recognizing and solving separable equations • Cross-Lesson: 5.01 Variation of Parameters

[First-Order Differential Equations, Separable Equations & Existence Theory · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Order and verifying solutions

A hot object cools from 100°C to 80°C in 10 minutes in an environment kept at 20°C. Under Newton's Law of Cooling, at what time t is the instantaneous rate of cooling $|dT/dt|$ exactly half of its initial value at $t = 0$?

a. $10\frac{\ln 2}{\ln(4/3)}$

b. $10\frac{\ln(4/3)}{\ln 2}$

c. 20 minutes

d. $5\frac{\ln 2}{\ln(4/3)}$

22 [Foundational] • Target: Recognizing and solving separable equations

An object cools in a room kept at a constant temperature T_a. At $t = 0$, its temperature is 100°C; at $t = 10$ min, it is 60°C; and at $t = 20$ min, it is 40°C. Assuming Newton's Law of Cooling, what is the room temperature T_a?

a. 20°C

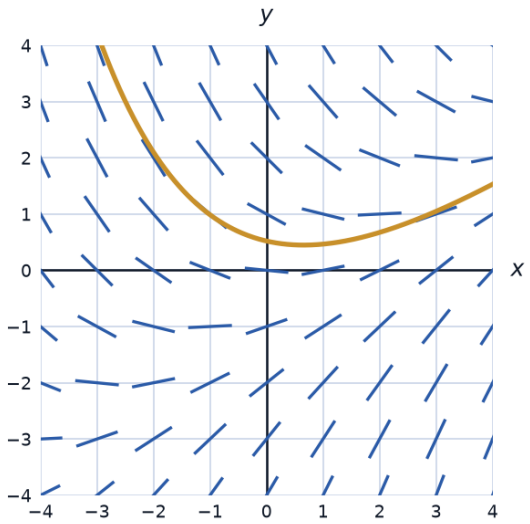
b. 15°C

c. 25°C

d. 18°C

23 [Foundational] • Target: Equilibrium solutions and qualitative behavior

Hot tea cools from 95°C to 80°C in 5 minutes in a room kept at 20°C. Under Newton's Law of Cooling, how many additional minutes will it take to cool from 80°C to 50°C?



a. $10\frac{\ln 2}{\ln(5/4)}$

b. $5\frac{\ln(4/5)}{\ln 2}$

c. $5\frac{\ln 2}{\ln(5/4)}$

d. $5\frac{\ln(5/2)}{\ln(5/4)}$

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24 [Foundational] • Target: General versus particular solutions

A metal ingot at $800\,^\circ\text{C}$ is immersed in a cooling bath kept at $50\,^\circ\text{C}$. After 1 minute, its temperature drops to $500\,^\circ\text{C}$. Under Newton's Law of Cooling, how many additional minutes will it take to cool to $100\,^\circ\text{C}$?

a. $\frac{3\ln 2}{\ln(5/3)}$

b. $\frac{\ln 3}{\ln(5/3)}$

c. $\frac{2\ln 3}{\ln(5/3)}$

d. $\frac{\ln 15}{\ln(5/3)}$

25 [Foundational] • Target: Initial value problems

An object cools in an environment whose ambient temperature rises linearly according to $T_m(t) = 10 + 2t$ (in $^\circ\text{C}$). Newton's Law of Cooling gives $T' = -k(T - T_m(t))$ with $k = 1$ hr^{-1}. If $T(0) = 50\,^\circ\text{C}$, what is $T(1)$?

a. $10 + 42e^{-1}$

b. $10 + 40e^{-1}$

c. $12 + 42e^{-1}$

d. $8 + 42e^{-1}$

26 [Foundational] • Target: Application: radioactive decay and population growth

A cup of coffee at $90\,^\circ\text{C}$ is placed in a room held at $20\,^\circ\text{C}$. After 10 minutes, the temperature of the coffee is $60\,^\circ\text{C}$. Assuming Newton's Law of Cooling, at what time t (in minutes) will its temperature reach $30\,^\circ\text{C}$?

a. $10\frac{\ln 7}{\ln(4/7)}$

b. $20\frac{\ln 7}{\ln(7/4)}$

c. $10\frac{\ln 7}{\ln(7/4)}$

d. $10\frac{\ln 4}{\ln(7/4)}$

27 [Foundational] • Target: Application: Newton's Law of Cooling

A thermometer initially reading $25\,^\circ\text{C}$ is taken outside into air at constant temperature T_m. After 1 minute it reads $15\,^\circ\text{C}$, and after 2 minutes it reads $9\,^\circ\text{C}$. Assuming Newton's Law of Cooling, what is the outdoor temperature T_m?

a. $-5\,^\circ\text{C}$

b. $5\,^\circ\text{C}$

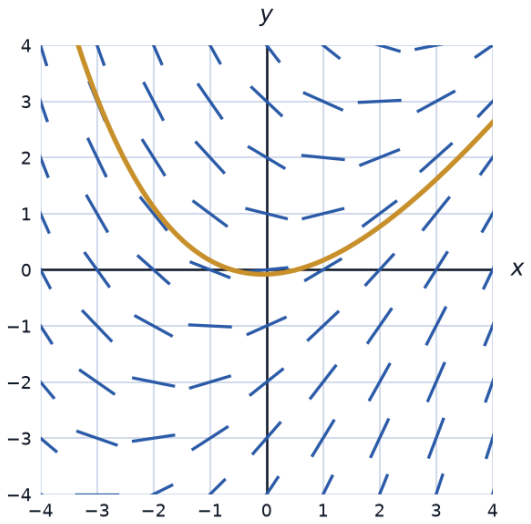
c. $3\,^\circ\text{C}$

d. $0\,^\circ\text{C}$

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28 [Foundational] • Target: Further practice: separable equations

Under Newton's Law of Cooling, $\frac{dT}{dt} = -k(T - T_m)$ with surrounding temperature $T_m = 20\,^\circ\text{C}$. An object starts at $T(0) = 80\,^\circ\text{C}$ and cools to $T(1) = 50\,^\circ\text{C}$ at $t = 1$ hour. What is the cooling constant k?



a. $-\ln 2$

b. $\ln 2$

c. $2\ln 2$

d. $\ln(5/8)$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Mixed review

Consider the differential equation $\frac{dy}{dx} = \frac{x^2\cos(y)}{1 - x^3}$ with the initial condition $y(0) = \frac{\pi}{2}$. If there exists a unique solution in a neighborhood of $x = 0$ and $y = \frac{\pi}{2}$, find the value of $y'(0)$.

a. 0

b. 1

c. $\frac{\pi}{2}$

d. Undefined

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30 [Intermediate] • Target: Order and verifying solutions

Consider the nonhomogeneous second-order linear differential equation $y'' - 5y' + 6y = 12x + 8$. $y_h(x) = c_1e^{2x} + c_2e^{3x}$ is the general solution to the corresponding homogeneous equation. Find the particular solution $y_p(x)$ using the method of undetermined coefficients, and then determine the value of the constant A such that $y_p(x) = Ax + B$ for some constant B.

a. $A = 2$

b. $A = 3$

c. $A = 6$

d. $A = 12$

31 [Intermediate] • Target: Recognizing and solving separable equations

Consider the second-order nonhomogeneous linear differential equation $y'' - 5y' + 6y = 12x + 4$. If a particular solution is of the form $y_p(x) = Ax + B$ and the general solution is $y(x) = C_1e^{2x} + C_2e^{3x} + Ax + B$, find the value of A.

a. 1

b. 2

c. 6

d. 12

32 [Intermediate] • Target: Equilibrium solutions and qualitative behavior

Consider the separable differential equation $\frac{dy}{dx} = 2x$ with the initial condition $y(0) = 3$. Find the value of $y(2)$.

a. 7

b. 6

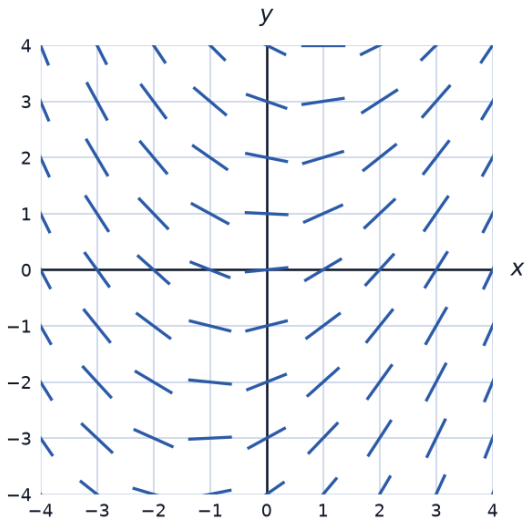
c. 8

d. 5

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33 **[Intermediate]** • **Target: General versus particular solutions**

A population grows according to $\frac{dP}{dt} = \frac{t}{1+t^2}P$ with initial size $P(0) = 500$. At what time $t > 0$ does the population reach 1300?



a. 2.6

b. $\sqrt{5}$

c. 3.2

d. 2.4

34 **[Intermediate]** • **Target: Initial value problems**

A population model exhibits accelerated growth satisfying $\frac{dP}{dt} = 0.02P^2$ with $P(0) = 50$. At what time t does the population experience finite-time blow-up (i.e., $P(t) \to \infty$)?

a. $t = 50$

b. $t = 1$

c. $t = 0.5$

d. $t = 2$

35 **[Intermediate]** • **Target: Application: radioactive decay and population growth**

A population of bacteria grows at a rate proportional to its size: $\frac{dP}{dt} = kP$. If the population triples every 4 hours, how many hours does it take to grow to 9 times its initial size?

a. 6

b. 16

c. 12

d. 8

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36 [Intermediate] • Target: Application: Newton's Law of Cooling

What is the general solution to the initial value problem $\frac{dy}{dt} = -3y$ with $y(0) = 50$?

a. $y(t) = 50 - e^{-3t}$

b. $y(t) = 50e^{3t}$

c. $y(t) = 50e^{-3t}$

d. $y(t) = -3t + 50$

37 [Intermediate] • Target: Further practice: separable equations

Consider the differential equation $\frac{dy}{dx} = kx(y^2 - 1)$, where k is a positive constant. If the solution passing through $(1, 2)$ also passes through $(2, 3)$, determine the value of the constant k.

a. $\frac{1}{4}\ln\left(\frac{3}{2}\right)$

b. $\frac{1}{3}\ln\left(\frac{2}{3}\right)$

c. $\frac{1}{4}\ln\left(\frac{2}{3}\right)$

d. $\frac{1}{3}\ln\left(\frac{3}{2}\right)$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Mixed review • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

Consider the second-order linear homogeneous differential equation $y'' - 2y' + y = 0$. \\ If one of the fundamental solutions is $y_1(x) = e^x$, and the Wronskian of two linearly independent solutions $y_1(x)$ and $y_2(x)$ is given by $W(y_1, y_2)(x) = 5e^{2x}$, find the unknown parameter k in $y_2(x) = kxe^x$.

a. $k = 5$

b. $k = -5$

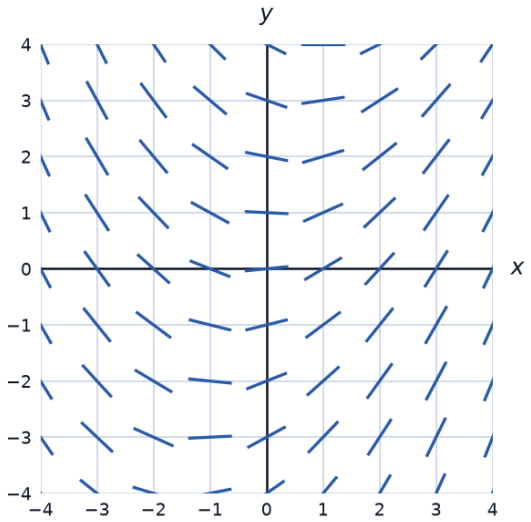
c. $k = 1$

d. $k = -1$

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39 [Challenging] • Target: Order and verifying solutions • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

Consider the separable differential equation $\frac{dy}{dx} = ky$ where k is a positive constant. A solution curve passes through the points $(0, 5)$ and $(2, 20)$. Determine the value of k and then find the value of y when $x = 3$.



a. $k = \ln 2, y(3) = 40$

b. $k = 2\ln 2, y(3) = 40$

c. $k = \ln 2, y(3) = 20\sqrt{2}$

d. $k = \ln 4, y(3) = 80$

40 [Challenging] • Target: Recognizing and solving separable equations • Cross-Lesson: 5.01 Variation of Parameters

Consider the separable first-order differential equation $\frac{dy}{dx} = \frac{x^2}{y}$. If the solution passes through the point $(1, 2)$, find the explicit solution for $y(x)$ assuming $y > 0$.

a. $y = \sqrt{\frac{x^3}{3} + \frac{7}{3}}$

b. $y = \sqrt{\frac{x^3}{3} + \frac{5}{3}}$

c. $y = \sqrt{\frac{x^3}{3} + 2}$

d. $y = \sqrt{\frac{x^3}{3} + \frac{2}{3}}$

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41 [Challenging] • Target: Equilibrium solutions and qualitative behavior • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Consider the second-order linear non-homogeneous differential equation: $\frac{d^2y}{dx^2} + 4y = \sec(2x)$. Suppose a particular solution is found using the method of variation of parameters. Let the fundamental set of solutions for the associated homogeneous equation be $y_1(x) = \cos(2x)$ and $y_2(x) = \sin(2x)$. If the particular solution is expressed in the form $y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x)$, determine the value of $u_2'(x)$.

a. $\frac{1}{2}$

b. $\frac{1}{4}$

c. $\frac{1}{2}\sin(2x)$

d. $\frac{1}{4}\cos(2x)$

42 [Challenging] • Target: General versus particular solutions • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

Consider the second-order nonhomogeneous linear differential equation $y'' - 2y' + y = e^x\sin(x)$. Let $y_p(x)$ be a particular solution obtained using the method of undetermined coefficients. Determine the value of $y_p''(\pi) - 2y_p'(\pi) + y_p(\pi)$.

a. 0

b. e^{π}

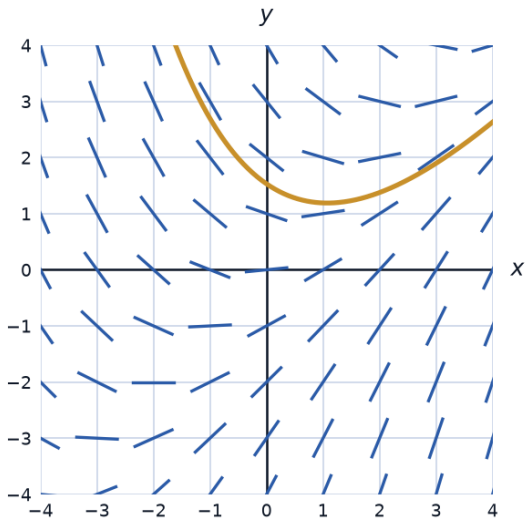
c. $-\pi e^{\pi}$

d. $e^{\pi}\sin(\pi)$

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43 **[Challenging]** • **Target: Initial value problems** • **Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients**

Consider a second-order linear non-homogeneous differential equation of the form $y''(x) + p(x)y'(x) + q(x)y(x) = g(x)$. If $y_1(x)$ and $y_2(x)$ form a fundamental set of solutions to the corresponding homogeneous equation $y''(x) + p(x)y'(x) + q(x)y(x) = 0$, and the Wronskian is $W(y_1, y_2)(x) = y_1(x)y_2'(x) - y_1'(x)y_2(x)$, which of the following expressions represents a particular solution $y_p(x)$ obtained using the method of variation of parameters?



a. $y_p(x) = -y_1(x)\int \frac{y_2(x)g(x)}{W(y_1,y_2)(x)}\,dx + y_2(x)\int \frac{y_1(x)g(x)}{W(y_1,y_2)(x)}\,dx$

b. $y_p(x) = y_1(x)\int \frac{y_2(x)g(x)}{W(y_1,y_2)(x)}\,dx + y_2(x)\int \frac{y_1(x)g(x)}{W(y_1,y_2)(x)}\,dx$

c. $y_p(x) = -y_1(x)\int \frac{y_2(x)g(x)}{W(y_1,y_2)(x)}\,dx - y_2(x)\int \frac{y_1(x)g(x)}{W(y_1,y_2)(x)}\,dx$

d. $y_p(x) = y_1(x)\int \frac{y_2(x)g(x)}{W(y_1,y_2)(x)}\,dx - y_2(x)\int \frac{y_1(x)g(x)}{W(y_1,y_2)(x)}\,dx$

44 **[Challenging]** • **Target: Application: radioactive decay and population growth** • **Cross-Lesson: 5.01 Variation of Parameters**

Consider the second-order linear homogeneous differential equation: $y''(x) - ky'(x) + 4y(x) = 0$. If the characteristic equation has a repeated real root, and one of the solutions to the differential equation is $y_1(x) = xe^{2x}$, determine the value of the parameter k.

a. 2

b. 4

c. 3

d. 1

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45 [Challenging] • Target: Application: Newton's Law of Cooling • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Consider the linear first-order differential equation $\frac{dy}{dx} + 2y = 6$. If the initial condition is $y(0) = 1$, find the value of $y(1)$.

a. 2

b. $e^2 - 1$

c. $3 - 2e^{-2}$

d. $3 - 2e^{-1}$

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